

# Appendix: The Bias– Variance Decomposition

INFO 521 · Module 2 · Lecture B — supplement

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Adapted with permission from lecture materials by Clayton Morrison, University of Arizona.

# Where we are

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Lecture B stated that, at a fixed input  $\mathbf{x}$ , expected test error splits into three pieces:

$$\mathbb{E}[(y - \hat{f}(\mathbf{x}))^2] = \text{bias}^2 + \text{variance} + \sigma^2.$$

This appendix derives that in full. Bookkeeping first — at the fixed  $\mathbf{x}$ :

- **Truth:**  $y = f(\mathbf{x}) + \varepsilon$ , with  $\mathbb{E}[\varepsilon] = 0$ ,  $\mathbb{E}[\varepsilon^2] = \sigma^2$ .
- **Estimate:**  $\hat{f} = \hat{f}_{\mathcal{D}}(\mathbf{x})$  — random *through the training set*  $\mathcal{D}$  (m3b's repeated-datasets picture, in prediction space).
- **Average fit:**  $\bar{f} = \mathbb{E}_{\mathcal{D}}[\hat{f}]$  — the orange curve from Lecture B's repeated-fits figure.
- $\varepsilon$  is a **new** test-time noise draw, independent of  $\mathcal{D}$ .

# The one trick: add and subtract $\bar{f}$

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Write the miss as three centered pieces:

$$y - \hat{f} = \underbrace{(f - \bar{f})}_{\text{systematic miss}} + \underbrace{(\bar{f} - \hat{f})}_{\text{dataset luck}} + \underbrace{\varepsilon}_{\text{fresh noise}}.$$

Square and take the expectation (over  $\mathcal{D}$  and  $\varepsilon$ ):

$$\mathbb{E}[(y - \hat{f})^2] = \mathbb{E}[(f - \bar{f})^2] + \mathbb{E}[(\bar{f} - \hat{f})^2] + \mathbb{E}[\varepsilon^2] + (\text{three cross terms}).$$

# The cross terms die

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Take each in turn:

$$\mathbb{E}[(f - \bar{f})(\bar{f} - \hat{f})] = (f - \bar{f}) \mathbb{E}_{\mathcal{D}}[\bar{f} - \hat{f}] = (f - \bar{f}) \cdot 0 = 0$$

—  $f - \bar{f}$  is a **constant** (no randomness left), and  $\hat{f}$  is centered on  $\bar{f}$  *by definition*.

$$\mathbb{E}[(f - \bar{f})\varepsilon] = (f - \bar{f}) \mathbb{E}[\varepsilon] = 0, \quad \mathbb{E}[(\bar{f} - \hat{f})\varepsilon] = \mathbb{E}_{\mathcal{D}}[\bar{f} - \hat{f}] \mathbb{E}[\varepsilon] = 0$$

— the test-time noise is mean-zero and **independent of the training set**, so both  $\varepsilon$  cross terms factor and vanish.

# The decomposition, earned

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What survives:

$$\mathbb{E}[(y - \hat{f}(\mathbf{x}))^2] = \underbrace{(f(\mathbf{x}) - \bar{f}(\mathbf{x}))^2}_{\text{bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}}[(\hat{f}(\mathbf{x}) - \bar{f}(\mathbf{x}))^2]}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible}}$$

- **bias<sup>2</sup>** — how far the *average* fit sits from the truth: rigidity's price.
- **variance** — how much a single fit swings around that average: flexibility's price.
- $\sigma^2$  — the noise floor; no model, however tuned, goes below it.

Complexity moves the first two in opposite directions; their sum is the U-curve of Lectures A/B.  
Ridge's  $\lambda$  walks you along it.

# Fine print

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Three things the derivation quietly fixed — say them out loud once:

- **One  $\mathbf{x}$  at a time.** The decomposition holds pointwise; “test error” in practice averages it over the input distribution.
- **Squared loss only.** The clean three-way split is a property of the square; other losses decompose differently (or not at all).
- **Expectations, not your dataset.** Bias and variance describe the *procedure* across repeated datasets — on your one sample you see a single draw, which is why CV (m2a) estimates rather than computes them.